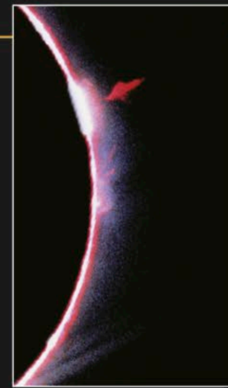
JOSEPH VON FRAUNHOFER

A German physicist and optical instrument maker, Joseph von Fraunhofer (1787–1826) is best known for his investigation of dark lines in the Sun's spectrum. Now known as Fraunhofer lines, they correspond to wavelengths of light absorbed by chemical elements in the outer parts of the Sun's atmosphere. Fraunhofer's observations were later used to help determine the composition of the Sun and other stars.

ATMOSPHERE

As well as forming its visible surface, the photosphere is the lowest layer of the Sun's atmosphere. Above it are three more atmospheric layers. The orangey-red chromosphere lies above the photosphere and is about 1,200 miles (2,000 km) deep. From the bottom to the top, its temperature rises from 8,100°F (4,500°C) to about 36,000°F (20,000°C). The chromosphere contains many flamelike columns of plasma called spicules, each rising up to 6,000 miles (10,000 km) high along local magnetic field lines and lasting for a few minutes. Between the chromosphere and the corona is a thin, irregular layer called the transition region, within which the temperature rises from 36,000°F (20,000°C) to about 1.8 million °F (1 million °C). Scientists are studying this region in an attempt to understand the cause of the temperature increase. The outermost layer of the solar atmosphere, the corona, consists of thin plasma. At a great distance from the Sun, this blends with the solar wind, a stream of charged particles (mainly protons and electrons) flowing away from the Sun across the solar system. The corona is extremely hot, 3.6 million °F (2 million °C), for reasons that are not entirely clear, although magnetic phenomena are believed to be a major cause of the heating. Coronal mass ejections (CMEs) are huge bubbles of plasma, containing billions of tons of material, that are occasionally ejected from the Sun's surface through the corona into space. CMEs can disturb the solar wind, which results in changes to aurorae in Earth's atmosphere (see p.74).

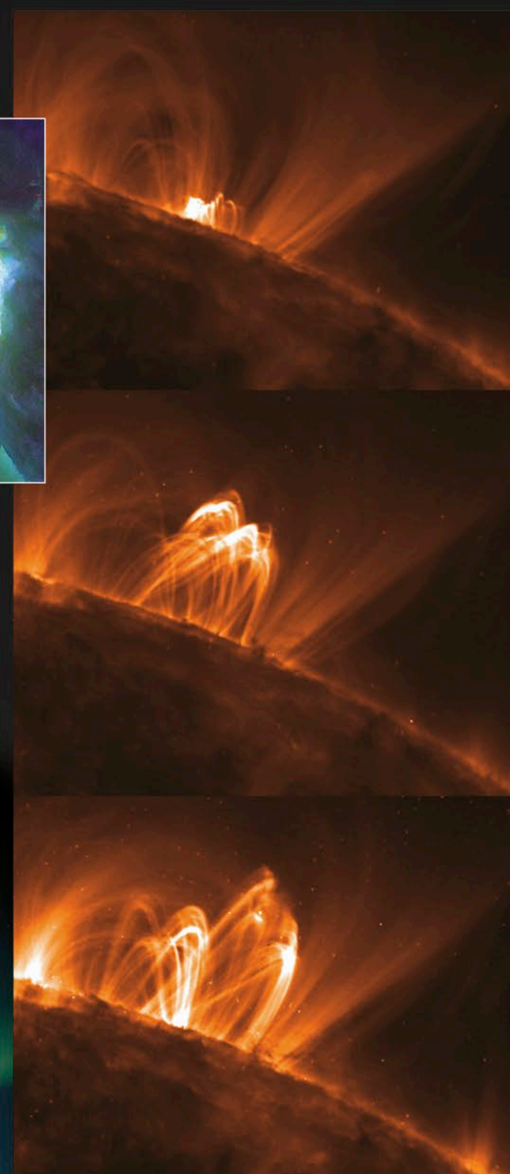


CHROMOSPHERE
The Sun's chromosphere is visible here as an irregular, thin red arc adjacent to the much brighter photosphere. Also apparent is a flamelike protuberance from the chromosphere into the corona.

CORONA
The outermost layer of the Sun, the corona extends outward into space for millions of miles from the chromosphere. It is most easily observed during a total eclipse of the Sun, as here.

CORONAL MASS EJECTION
This image of a coronal mass ejection (top left) was taken by the SOHO solar observatory using a coronagraph—an instrument that blocks direct sunlight by means of an occulter (the central smooth red area in the image). The white circle represents the occulted disk of the Sun.

MAGNETIC ERUPTION
Hot plasma explodes into the atmosphere, following magnetic field lines. In this TRACE image, colors represent temperature, with blue being the coolest and red the hottest.



POSTFLARE LOOPS
These three images of a magnetically active solar region, taken by the TRACE satellite, span a period of 2.5 hours. The loops in the Sun's corona probably followed a solar flare and consist of plasma heated to exceedingly high temperatures along magnetic field lines.

NORTHERN LIGHTS
When charged particles from the solar wind reach Earth, they can cause aurorae. This photograph of the aurora borealis was taken in Manitoba, Canada.